

GE261004 DESIGN THINKING

Unit II – Empathy and Problem Definition · Presentation 5 of 6

Problem Identification and Framing

From field notes to insight · what makes a frame work · the anatomy of a problem statement · weak and strong statements · How Might We questions

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In this presentation

- What you have after the field
- Synthesis: from notes to insight
- Three ways a frame goes wrong
- The anatomy of a statement
- Weak and strong statements
- How Might We questions
- HMW: too tight, too loose
- In class: framing your problem

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From findings to a problem

The hardest and least taught step in the whole process

What you have after the field

- Pages of notes, quotations, photographs and half-formed impressions
- Several contradictions, at least one surprise, and no obvious answer
- This is the correct state to be in – it means you gathered rather than confirmed
- The work now is synthesis: turning many observations into a few statements

KEY POINT If the field produced no surprise, you were collecting evidence for a conclusion you had already reached.

Synthesis: from notes to insight

- Put every observation on its own note – one idea per note
- Cluster by what they have in common, not by which interview they came from
- Name each cluster with a sentence: a word is a category, a sentence is a finding
- The insight is the “because” – the reason behind behaviour you did not expect

KEY POINT An observation says what happened. An insight says why, and lets you predict what would happen next.

Too narrow, too broad, well framed

TOO NARROW

"Design a louder library door buzzer"

the answer is already hidden in the question

WELL FRAMED

"How might we help students find quiet study space?"

one clear user and need, many possible answers

TOO BROAD

"Improve the student experience"

true of everything, so it guides nothing

narrow enough to act on, wide enough to allow many answers

The same situation, framed three ways – only one of them is designable.

Three ways a frame goes wrong

Too narrow

The solution is already inside the question, so there is nothing left to design.

Too broad

It is true of everything, so it rules nothing out and guides no decision.

Solution-shaped

It names a technology instead of a need, and locks you in before you begin.

KEY POINT Judge a frame by the answers it allows, not by how impressive it sounds.

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Writing the statement

User, need and insight – and nothing else

The anatomy of a problem statement

- Names one user specifically – not “people”, not “students in general”
- States a need as something they must be able to DO, not a thing they must GET
- Gives the insight: why the need exists and why it is not already met
- Contains no solution, no technology and no brand name

KEY POINT User + need + insight. If any one of the three is missing, the statement will not survive ideation.

Weak and strong statements

Weak	What is wrong	Stronger
“Build a library booking app”	Solution baked in	Students need to know a seat is free before walking over
“Improve campus facilities”	Guides nothing	Commuting students need somewhere to work between classes
“Users want a better interface”	No user, no insight	First-time users need to finish without asking anyone

How Might We questions

- Turn the problem statement into an invitation: “How might we...”
- **How** assumes a way exists; **might** allows many; **we** makes it the team's job
- Write five or six from one statement, at different levels of ambition
- Keep them free of solutions – “How might we build an app” is not one

KEY POINT A good How Might We makes you want to start writing ideas immediately. If it does not, rewrite it.

How Might We: too tight, too loose

Too tight

- How might we add a buzzer?
- How might we build an app?
- How might we print a notice?
- Only one answer fits

Too loose

- How might we fix education?
- How might we improve life?
- How might we redesign campus?
- Any answer at all fits

KEY POINT Aim between the two: several genuinely different answers should be possible, and you should be able to start today.

In class: framing your problem

- Cluster your team's field notes and name each cluster in a full sentence
- Write one problem statement: user, need, insight – in that order
- Generate six How Might We questions from it, at different scales
- Pick the one that makes the team most impatient to start

ACTIVITY Twenty-five minutes. Every team pins up one problem statement and six How Might We questions before the hour ends.

THINK ABOUT IT

Read your problem statement to someone from another team. Can they start designing without asking you anything?

- Which word made them pause?
- Did they hear a user, a need and an insight?
- Could they name three different directions from it?
- What would you cut to make it sharper?

Summary

- Field work ends in mess and surprise – that is the correct state
- Synthesis clusters observations and names each cluster in a sentence
- An insight is the “because” behind unexpected behaviour
- A frame fails by being too narrow, too broad or solution-shaped
- A statement carries one user, a verb-shaped need and an insight
- How Might We turns the statement into an invitation to ideate

Next: Point of view statements and reframing

- The POV template and its three parts
- Why needs are verbs and never nouns
- Testing whether a POV is finished
- Four ways to reframe, and when to do it